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Image-based barcode detection

Abstract

A method includes: capturing an image; partitioning the image into sub-images; for each sub-image: providing the sub-image to a detection model, receiving, from the detection model, one or more sub-image regions of interest (SROIs), each SROI defined by (i) a position of the SROI, and (ii) one of a set of symbology categories, each symbology category encompassing a plurality of barcode symbologies; generating one or more regions of interest (ROI) from the SROIs, each ROI defined by a position of the ROI in the image, and a symbology category; and providing the ROIs to a decoder.

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Background/Summary

BACKGROUND

(1) Images captured by digital image sensors can be used to detect and decode barcodes. Detecting and decoding barcodes from an image with a field of view that encompasses multiple barcodes, e.g., displayed on objects on a shelf or the like, may be complicated by the limited portion of the image representing each barcode. Capturing such an image at a higher resolution may improve detection and decoding accuracy, but may also increase the computational burden of detection and decoding.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

- (1) The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, together with the detailed description below, are incorporated in and form part of the specification, and serve to further illustrate embodiments of concepts that include the claimed invention and explain various principles and advantages of those embodiments.
- (2) FIG. 1 is a diagram of a computing device for image-based barcode detection.
- (3) FIG. 2 is a flowchart of a method of image-based barcode detection.

- (4) FIG. 3 is a diagram illustrating an example performance of blocks **205** and **210** of the method of FIG. 2.
- (5) FIG. 4 is a diagram illustrating example performances of blocks **215** and **215** of the method of FIG. 2.
- (6) FIG. 5 is a diagram illustrating an example performance of block **230** of the method of FIG. 2.
- (7) FIG. 6 is a diagram illustrating an example performance of block **240** of the method of FIG. 2.
- (8) FIG. 7 is a diagram illustrating an example performance of block **255** of the method of FIG. 2.
- (9) Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of embodiments of the present invention.
- (10) The apparatus and method components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present invention so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

DETAILED DESCRIPTION

- (11) Examples disclosed herein are directed to a method including: capturing an image; partitioning the image into sub-images; for each sub-image: providing the sub-image to a detection model, receiving, from the detection model, one or more sub-image regions of interest (SROIs), each SROI defined by (i) a position of the SROI, and (ii) one of a set of symbology categories, each symbology category encompassing a plurality of barcode symbologies; generating one or more regions of interest (ROI) from the SROIs, each ROI defined by a position of the ROI in the image, and a symbology category; and providing the ROIs to a decoder.
- (12) Additional examples disclosed herein are directed to a computing device, comprising: an image sensor; and a processor configured to: capture an image via the image sensor; partition the image into sub-images; for each sub-image: provide the sub-image to a detection model, receive, from the detection model, one or more sub-image regions of interest (SROIs), each SROI defined by (i) a position of the SROI, and (ii) one of a set of symbology categories, each symbology category encompassing a plurality of barcode symbologies; generate one or more regions of interest (ROI) from the SROIs, each ROI defined by a position of the ROI in the image, and a symbology category; and provide the ROIs to a decoder.
- (13) FIG. 1 illustrates a computing device **100**, such as a mobile computer, a smart phone, a barcode scanner, or the like. The device **100** includes a housing supporting various components of the device **100**, discussed below. The device **100** can be operated to capture images, and is configured to detect barcodes within such images, and under certain conditions, to decode the detected barcodes. In the discussion below, the term “barcode” includes symbols using one-dimensional symbologies that encode data horizontally (e.g., UPC-A, Code **128**, and the like), and symbols with two-dimensional symbologies (e.g., Data Matrix, QR Code, DotCode, and the like).
- (14) For example, the device **100** can be operated to capture an image representing a region **104** of a shelf or other support structure **108** that supports one or more barcodes **112** (illustrated as grey boxes in FIG. 1). In the illustrated example, the support structure **108** includes shelves, and the barcodes **112** are disposed on shelf edges, e.g., facing into an aisle of a facility in which the device **100** is deployed. A wide variety of other objects, or collections of objects, can carry the barcodes **112** in other examples. For example, the barcodes **112** can be affixed to packages in a sorting facility, on products on the support structure **108**, or the like.
- (15) The region **104** of the support structure **108** contains six barcodes **112** in the illustrated example. In a single image depicting the region **104**, each barcode **112** may occupy only a small portion of the image. For example, in the illustrated example each barcode **112** may occupy only about two percent of the area of the region **104**. At certain image resolutions (e.g., images having

pixel counts lower than about one megapixel, or 1 MP), an image depicting a relatively large field of view (FOV) such as that encompassing the region **104** may yield inaccurate or incomplete barcode detection and/or decoding results, as a result of the small number of pixels representing each barcode **112**.

(16) The accuracy of the detection and decoding process may be improved, for example, by capturing images with greater resolution (e.g., with a pixel count greater than about 10 MP). However, larger images also impose a greater computational burden on the device **100**. Under some conditions, e.g., when the device **100** is a mobile computer or handheld barcode scanner, the computational load associated with detecting and decoding barcodes **112** in a large image (e.g., an image with a pixel count greater than about 10 MP) may introduce delays to the detection and decoding process (e.g., delays greater than about five seconds) sufficient to impede downstream processes, such as product picking, package transport, or the like. The device **100** therefore implements certain functionality, discussed below, to mitigate the performance impact of using large images for barcode detection and decoding.

(17) Certain internal components of the device **100** are shown in FIG. **1**. The device **100** includes a processor **120**, such as a central processing unit (CPU), graphics processing unit (GPU), application-specific integrated circuit (ASIC), or the like. The processor **120** is communicatively coupled with a non-transitory computer-readable storage medium such as a memory **124**, e.g., a combination of volatile memory elements (e.g., random access memory (RAM)) and non-volatile memory elements (e.g., flash memory or the like). The memory **124** stores a plurality of computer-readable instructions in the form of applications, including in the illustrated example a barcode detection application **128**, whose execution by the processor **120** configures the device **100** to process the images to detect and/or decode barcodes **112** therein.

(18) The device **100** can also include a communications interface **132**, enabling the device **100** to communicate with other computing devices (not shown) via any suitable communications links. The device **100** further includes a sensor **136**, such as a camera or other suitable image sensor, configured to capture images representing a portion of the surroundings of the device **100** within a FOV **140** of the sensor **136**. As discussed below, execution of the application **128** configures the processor **120** to process such images to detect and/or decode barcodes **112** in the images captured by the sensor **136**. The device **100** can also include one or more output devices, such as a display **144**, a speaker (not shown), or the like. In other examples, the display **144** can be omitted, e.g., in the case of a barcode scanner with a ring form factor. The device **100** can further include one or more input devices, such as a microphone, a touch screen (e.g., integrated with the display **144**), a keypad, a scan trigger, or the like.

(19) The device **100** implements, for example via the application **128**, an image pre-processor **148** configured to process images captured via the sensor **136**, e.g., to partition the images into sub-images as described below. The device **100** also implements a detector **152** configured to process the sub-images produced by the pre-processor **148** to detect regions of interest therein containing barcodes, and a decoder **156** configured to process the regions of interest from the detector **152** to decode the barcodes contained in the regions of interest. In the illustrated example, the pre-processor **148**, the detector **152**, and the decoder **156** are implemented by the processor **120** via execution of the application **128**. In other examples, however, one or more of the pre-processor **148**, the detector **152**, and the decoder **156** can be implemented in distinct hardware elements, via separate applications, and the like. For example, the processor **120** can implement the pre-processor **148** and the detector **152**, e.g., via execution of the application **128**, and the device **100** can include an additional controller or other processing hardware implementing the decoder **156**, such as a scan engine. In further examples, the processor **120** can implement the detector **152** and the decoder **156** via execution of the application **128**, while the pre-processor **148** can be implemented via an additional controller or other processing hardware, e.g., integrated with the sensor **136**. In still further examples, the pre-processor **148** and the decoder **156** can be implemented by the processor

120, while the detector **152** is implemented by a separate controller.

(20) Turning to FIG. 2, a method **200** of detecting and decoding barcodes in images is illustrated. The method **200** is described below in conjunction with its performance at the device **100**, e.g., via execution of the application **128** by the processor **120**.

(21) At block **205**, the device **100** is configured to capture an image, corresponding to a portion of the surroundings of the device **100** within the FOV **140** of the sensor **136**. For example, the device **100** can be operated to capture an image of a portion of the support structure **108** as shown in FIG. 1. A wide variety of objects may be positioned in the FOV **140** when the image is captured, in addition to or instead of the support structure **108**. The device **100** can capture the image at block **205** in response to activation of an input (e.g., a trigger, a button, or the like). In other examples, the device **100** can be configured to capture a sequence of images, e.g., a video stream. Following initiation of the video stream capture, e.g., via an input as set out above, each image is captured automatically, without further input activation, at a suitable capture frequency (e.g., ten frames per second, although lower and higher frame rates are also contemplated).

(22) At block **210**, the device **100** is configured to partition the image captured at block **205** into a plurality of sub-images. The sub-images, as discussed below, are processed independently to detect barcodes or portions of barcodes therein, and the detection results are subsequently combined to represent barcode detections for the initial image from block **205**. Independently processing the sub-images can reduce the computational load of detecting barcodes in the initial image. That is, the computational burden associated with partitioning the image, processing each sub-image, and combining the results of such processing, can be smaller than the computational burden associated with performing detection on the initial image as a whole, facilitating the use of high-resolution images for real-time or near real-time barcode detection (e.g., completing detection over the whole initial image within about five seconds).

(23) Partitioning the image into sub-images can be performed using various mechanisms. For example, the device **100** can be configured to maintain a predetermined target sub-image size (e.g., expressed as a pixel count). The target size can be defined in the application **128**, as a configuration setting, or the like. The device **100** can compare the target size to the size of the image from block **205**, and determine a number of sub-images into which to partition the image such that each sub-image has a size equal to or smaller than the target. For example, for an initial image with a 16 MP pixel count, and a target sub-image size of 1 MP, the device **100** can determine that the image is to be partitioned into sixteen equally-sized sub-images.

(24) The device **100** can further determine the dimensions (e.g., a width and a height, measured in pixels) of each sub-image, based on the size of the initial image and the number of sub-images. For example, the device **100** can factorize the number of sub-images, e.g., yielding one or more pairs of factors. The device **100** can then select the pair of factors with the smallest difference between the pair. For example, if the number of sub-images is sixteen, the factors [4, 4] may be selected rather than the factors [8, 2]. The dimensions of the initial image can then be divided by the selected factors. Various other mechanisms for determine the number and size of the sub-images can also be employed. For example, the sensor **136** may operate at a fixed resolution, and the device **100** can maintain a fixed number of sub-images, as well as fixed dimensions for the sub-images, determined prior to performance of the method **200** and stored as configuration data.

(25) Turning to FIG. 3, an example performance of blocks **205** and **210** of the method **200** is illustrated. At block **205**, the device **100** is configured to capture, via the sensor **136**, an image **300** of the support structure **108**, depicting barcodes **112-1**, **112-2**, **112-3**, **112-4**, **112-5**, and **112-6** (collectively referred to herein as the barcodes **112**, and generically referred to as a barcode **112**; similar nomenclature is also used herein for other components with suffixed reference numbers). As will be apparent from FIG. 3, the barcodes **112** use different symbologies. For example, the barcodes **112-1**, **112-4**, and **112-5** use two-dimensional symbologies, while the barcodes **112-2**, **112-3**, and **112-6** use one-dimensional symbologies.

(26) The image **300** has a size defined by a width **304** and a height **308**, e.g., in pixels. The width **304** and height **308** can be defined by the resolution of the sensor **136**. To determine the number of sub-images into which the image **300** is to be partitioned, the device **100** can divide the pixel count (e.g., the width **304** times the height **308**) by a target sub-image pixel count, e.g., rounding up to the nearest integer. In some examples, the device **100** can be configured to round the result of the above division up to the nearest even integer, to avoid determining a prime number of sub-images.

(27) In the present example, the target pixel count is about one eighth of the pixel count of the image **300**. For example, the size of the image **300** can be about 16 MP, and the target pixel count can be about 2 MP. A wide variety of other image sizes and target sub-image sizes are contemplated. In this example, the device **100** therefore determines at block **210** that the image **300** is to be partitioned into eight sub-images of equal size. The device **100** can further determine the dimensions (e.g., width and height) of each sub-image by factorizing the sub-image count, and selecting the closest pair of factors. The pairs of factors in this example are [1, 8] and [2, 4], and the device **100** can therefore be configured to select the latter pair, and to partition the image **300** into an array two sub-images wide and four sub-images high. Eight sub-images **312-1**, **312-2**, **312-3**, **312-4**, **312-5**, **312-6**, **312-7**, and **312-8** are shown in FIG. 3. Each sub-image **312** has a width equal to one half of the width **304**, and a height equal to one quarter of the height **308**. The positions of the sub-images **312** relative to the image **300** are stored, such that positions of barcodes derived from the sub-images can be mapped to positions within the image **300**. The above partitioning operations can divide an arbitrarily sized initial image into a suitable number of sub-images, according to a target sub-image size.

(28) Returning to FIG. 2, at block **215** the device **100** is configured to provide the sub-images independently to a detection model, such as a convolutional neural network or other suitable model. In some examples, the detection model can be based on a version of the You Only Look Once (YOLO) model, e.g., YOLO v5. The model can be trained, prior to performance of the method **200**, to detect barcodes in input images, and to output the position of each detected barcode in the input image, as well as a class of the detected barcode. The class indicates a symbology category. For example, the model can be trained to assign one of a 2D category (indicating that the detected barcode uses a two-dimensional symbology) and a 1D category (indicating that the detected barcode uses a one-dimensional symbology). In other examples, additional categories can be implemented, including categories encompassing a single symbology. For example, in some embodiments the model can assign to each detected barcode one of the 2D category, the 1D category, and a DotCode category indicating that the detected barcode uses the DotCode symbology specifically.

(29) As will be apparent, the model can be trained by providing a set of labelled input images, with each input image annotated with the positions of barcodes in the image and symbology categories for the barcodes. The training image set can also include partial barcodes, to facilitate detection of partial barcodes in the sub-images from block **210**, as discussed below.

(30) At block **215**, the device **100** is configured to select a sub-image **312** that has not yet been provided to the model, and to provide the sub-image **312** to the model. At block **220**, the device **100** is configured to obtain, as output from the model, zero or more sub-image regions of interest (SROIs). Each SROI indicates that the model detected a barcode or partial barcode, and is defined by a position within the sub-image and a symbology category. The position can be expressed, for example, as the pixel coordinates of a top-left corner of the SROI, as well as a height and width of the SROI (also in pixels). The position defining an SROI can take various other forms, including for example the pixel coordinates of opposing corners (e.g., top-left and bottom-right corners), pixel coordinates of all four corners of the SROI, or the like.

(31) In the present example, the model is implemented at the device **100**, e.g., as a component of the application **128**. For example, the trained model can be deployed to the device **100** prior to performance of the method **200** using a suitable software library, such as TensorFlow. In some

examples, the device **100** can be configured to transmit the sub-images to another computing device for processing, e.g., via the communications interface **132**. The other computing device can implement the model, and return the detection results to the device **100**.

(32) At block **225**, having obtained detection results from the model for a sub-image **312**, the device **100** is configured to determine whether any sub-images **312** remain to be processed via the model. When the determination at block **225** is affirmative, the device **100** repeats block **215** with the next sub-image, until each of the sub-images **312** has been provided to the model. When the determination at block **225** is negative, indicating that all the sub-images generated at block **210** have been processed via the model, performance of the method **200** proceeds to block **230**.

(33) FIG. **4** illustrates example performances of block **220** for the sub-images **312-3** and **312-7**. In particular, by processing the sub-image **312-3** via the detection model, the device **100** obtains two SROIs **400** and **404**. The SROI **400** corresponds to the barcode **112-3**, and the SROI **404** corresponds to a portion of the barcode **112-4** (which was divided between the images **312-3** and **312-7**, as seen in FIG. **3**). Further, by processing the sub-image **312-7** via the detection model, the device **100** obtains another SROI **408**, corresponding to another portion of the barcode **112-4**. Each SROI is defined by a position, e.g., expressed as the coordinates of a top-left corner of a bounding box (x1, y1, etc.) and dimensions (e.g., W1, D1, etc.). Each SROI is further defined by a symbology category (e.g., 1D or 2D), which can be associated with a confidence level indicating a probability that the symbology category was correctly detected. In some examples, SROIs with confidence levels below a threshold (e.g., 50% or any other suitable predetermined threshold) can be discarded.

(34) Returning to FIG. **2**, at block **230** the device **100** is configured to generate regions of interest (ROIs) from the SROIs obtained via successive performances of blocks **215** and **220** (one performance of blocks **215** and **220** per sub-image generated at block **210**). Each ROI is defined by a position of the ROI within the image **300** (e.g., corner coordinates and pixel dimensions), as well as a symbology category. To generate the ROIs at block **230**, the device **100** can be configured to determine whether any of the SROIs are contiguous. For an SROI that is not contiguous with any other SROIs, the device **100** converts the position defining the SROI, which corresponds to a position in a sub-image **312**, to a position in the image **300**. For an SROI that is contiguous with one or more other SROIs, the device **100** is configured to merge the contiguous SROIs to generate an ROI, as discussed below in connection with FIG. **5**.

(35) Referring to FIG. **5**, the SROIs shown in FIG. **4** are shown superimposed on the boundaries of the image **300**. The positions of the sub-images **312** within the image **300** are maintained, as noted earlier, and a position within a given sub-image **312** can therefore be converted to a position within the image **300**. To determine whether any SROIs are contiguous, the device **100** can be configured to determine a distance between a pair of SROIs (e.g., the shortest distance between the pair of SROIs), such as a distance **500** between the SROIs **404** and **408**, shown in the lower-left detail view of FIG. **5**. When the distance **500** is below a predetermined threshold (e.g., a number of pixels, a fraction of a width or height of either of the SROIs, or the like), the SROIs **404** and **408** are considered contiguous.

(36) When a pair of SROIs is considered contiguous, the device **100** is configured to generate an ROI by merging the contiguous SROIs. Merging SROIs can be implemented by, for example, extending the width of one SROI of the pair to encompass the width of the other SROI, and extending the height of the one SROI to encompass the height of the other SROI. FIG. **5** illustrates an ROI **504** resulting from the merger of the SROIs **404** and **408**. The merged ROI **504** can include the same symbology category as the contributing SROIs. In some examples, SROIs are only considered contiguous if they are sufficiently close to one another and also have matching symbology categories. The merging process set out above can be repeated until no further mergers are performed. In some examples, more than two SROIs can therefore contribute to one ROI. Merged ROIs such as the ROI **504** can result from SROIs that are side by side, as shown in FIG. **5**,

or arranged vertically relative to one another, or a combination thereof.

(37) Returning to FIG. 2, at block **235** the device **100** is configured to provide the ROIs from block **230** to a decoder, e.g., implemented via the application **128**, or as a separate application at the device **100**. The decoder can include an executable process, dedicated hardware, or the like, configured to determine the symbology of a barcode and decode data from the barcode based on the determined symbology. The output of the decoder can include, for example, a decoded string and a confidence level, or an indication that decoding was not successful (e.g., some barcodes **112** may be represented in sufficient detail in the image **300** to be detected but not decoded).

(38) Providing an ROI to the decoder includes providing the image **300** and the position of the ROI, and/or an extracted portion of the image **300** corresponding to the position of the ROI. Providing an ROI to the decoder also includes providing the symbology category associated with the ROI. The symbology category facilitates the determination of a specific symbology for the ROI by the decoder, by reducing the number of candidate symbologies to attempt. For example, an ROI provided to the decoder with the “2D” symbology category allows the decoder to ignore one-dimensional symbologies in searching for the symbology of the ROI, and attempt to decode the ROI using each two-dimensional symbology.

(39) At block **240**, having provided each of the ROIs (in sequence, or in parallel, according to the capabilities of the decoder), the device **100** is configured to obtain decode results, and can be configured to display the ROIs along with visual indicators of decode status, based on the decode results. For example, the device **100** can control the display **144** to present the image **300**, with each ROI highlighted in a first color, pattern, or the like if the ROI was successfully decoded, or in a second color, pattern, or the like if the ROI was not successfully decoded. The data decoded from each successfully decoded ROI can also be presented on the display **144**, in some examples.

(40) Turning to FIG. 6, an example performance of block **240** is illustrated, in which the image **300** is presented (e.g., on the display **144**, not shown), with visual indicators **600** associated with successfully decoded ROIs, and visual indicators **604** associated with unsuccessfully decoded ROIs. In particular, as shown in FIG. 6, the ROI encompassing the barcode **112-2** was not successfully decoded, and that ROI is therefore presented with a visual indicator **604**, visually distinct from the visual indicators **600** associated with the remaining ROIs. The visual indicators **600** and **604** can indicate to an operator of the device **100** which barcodes **112** may require further images (e.g., captured closer to the barcode **112**) to decode. The device **100** can also, in some examples, compare any decoded data to a pick list or the like, and present an additional visual indicator in association with ROIs that match the pick list, e.g., to guide the operator to items to be picked from a shelf or the like.

(41) Referring again to FIG. 2, in some examples, the device **100** can return from block **240** to block **205**, to capture a further image and repeat the detection and decoding process set out above. In the present example, at block **245**, the device **100** can further determine three-dimensional locations of the ROIs presented at block **240**, relative to the device **100**. For example, the device can include a motion sensor, such as an inertial measurement unit (IMU) or the like, and can employ the motion sensor and/or the image sensor **136** to track a pose (e.g., a position and orientation of the device **100** in three-dimensional space). Pose tracking using motion sensor and/or image data can be implemented using various mechanisms, including ARCore (Google™) or ARKit (Apple™).

(42) Based on the pose of the device **100** and the position of an ROI in the image **300**, the device **100** can determine a position in three dimensions of the ROI, relative to the device. The device **100** can maintain a count of distinct images in which an ROI has been detected at the same three-dimensional position (e.g., within a threshold distance of the three-dimensional position). At block **250**, the device **100** can be configured to determine whether any ROIs have been detected a threshold number of times (e.g., in a threshold number of distinct images), without being successfully decoded. When the determination at block **250** is negative, the device **100** can return

to block 205. When the determination at block 250 is affirmative, the device 100 can update the visual indicator associated with the ROI at block 255. The visual indicator can be changed to a third color, for example, and/or to a different shape, size, or the like. Turning to FIG. 7, for example, if the barcode 112-2 is detected a threshold number of times (e.g., in five successive image frames), and has not been decoded, the device 100 can replace the indicator 604 with an indicator 700, to draw attention to the barcode 112-2.

(43) In other examples, the detection functionality discussed above can be applied to other forms of machine readable indicia than barcodes. For example, the device 100 can be configured to detect and categorize indicia such as text (e.g., as part of an optical character recognition (OCR) pipeline). In such examples, the detector 152 can identify sections of text, e.g., paragraphs or the like, and select an indicia category instead of the symbology category mentioned above. The indicia category can correspond to a plurality of languages (e.g., Latin-based languages, Slavic languages, and the like), a plurality of textual fonts, and the like.

(44) In the foregoing specification, specific embodiments have been described. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the invention as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of present teachings.

(45) The benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims. The invention is defined solely by the appended claims including any amendments made during the pendency of this application and all equivalents of those claims as issued.

(46) Moreover in this document, relational terms such as first and second, top and bottom, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” “has,” “having,” “includes,” “including,” “contains,” “containing” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises, has, includes, contains a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a”, “has . . . a”, “includes . . . a”, “contains . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises, has, includes, contains the element. The terms “a” and “an” are defined as one or more unless explicitly stated otherwise herein. The terms “substantially”, “essentially”, “approximately”, “about” or any other version thereof, are defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting embodiment the term is defined to be within 10%, in another embodiment within 5%, in another embodiment within 1% and in another embodiment within 0.5%. The term “coupled” as used herein is defined as connected, although not necessarily directly and not necessarily mechanically. A device or structure that is “configured” in a certain way is configured in at least that way, but may also be configured in ways that are not listed.

(47) Certain expressions may be employed herein to list combinations of elements. Examples of such expressions include: “at least one of A, B, and C”; “one or more of A, B, and C”; “at least one of A, B, or C”; “one or more of A, B, or C”. Unless expressly indicated otherwise, the above expressions encompass any combination of A and/or B and/or C.

(48) It will be appreciated that some embodiments may be comprised of one or more specialized processors (or “processing devices”) such as microprocessors, digital signal processors, customized processors and field programmable gate arrays (FPGAs) and unique stored program instructions (including both software and firmware) that control the one or more processors to implement, in

conjunction with certain non-processor circuits, some, most, or all of the functions of the method and/or apparatus described herein. Alternatively, some or all functions could be implemented by a state machine that has no stored program instructions, or in one or more application specific integrated circuits (ASICs), in which each function or some combinations of certain of the functions are implemented as custom logic. Of course, a combination of the two approaches could be used.

(49) Moreover, an embodiment can be implemented as a computer-readable storage medium having computer readable code stored thereon for programming a computer (e.g., comprising a processor) to perform a method as described and claimed herein. Examples of such computer-readable storage mediums include, but are not limited to, a hard disk, a CD-ROM, an optical storage device, a magnetic storage device, a ROM (Read Only Memory), a PROM (Programmable Read Only Memory), an EPROM (Erasable Programmable Read Only Memory), an EEPROM (Electrically Erasable Programmable Read Only Memory) and a Flash memory. Further, it is expected that one of ordinary skill, notwithstanding possibly significant effort and many design choices motivated by, for example, available time, current technology, and economic considerations, when guided by the concepts and principles disclosed herein will be readily capable of generating such software instructions and programs and ICs with minimal experimentation.

(50) The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

Claims

1. A method, comprising: capturing an image; partitioning the image into sub-images; for a sub-image: providing the sub-image to a detector executing a detection model, receiving, from the detector, a sub-image region of interest (SROI) defined by (i) a position of the SROI within the sub-image, and (ii) a symbology category encompassing a plurality of barcode symbologies; generating a region of interest (ROI) from the SROI by converting the position of the SROI in the sub-image to a position in the image, the ROI defined by the position in the image and the symbology category; providing the ROI to a barcode decoder; and receiving, from the detector, a further SROI having a position within a further sub-image, wherein: generating the ROI from the SROI comprises: identifying SROIs with contiguous positions in the image, and merging the SROIs with contiguous positions.
2. The method of claim 1, further comprising, prior to partitioning the image into the sub-images: determining a number of the sub-images based on a target size of the sub-image.
3. The method of claim 2, wherein the target size of the sub-image defines a pixel count; and wherein determining the number of the sub-images comprises: dividing a pixel count of the image by the pixel count defined by the target size of the sub-image.
4. The method of claim 1, wherein the symbology category includes one of a first category corresponding to one-dimensional barcodes, and a second category corresponding to two-dimensional barcodes.
5. The method of claim 1, further comprising, at the barcode decoder: selecting a set of barcode symbologies according to the symbology category corresponding to the ROI; determining which of

the symbologies matches the ROI; and decoding data from the ROI based on the symbology matching the ROI.

6. The method of claim 5, further comprising: controlling a display to present the position of the ROI and a visual indicator of a decoding status; determining a three-dimensional location corresponding to the ROI; determining, based on the three-dimensional location, that the ROI was detected in a threshold number of previous images and has not been decoded; and updating the visual indicator associated with the ROI.

7. A computing device, comprising: an image sensor; and a image pre-processor configured to: capture an image via the image sensor; partition the image into sub-images; and provide the sub-images to a detector; a detector configured to, for a sub-image: execute a detection model to obtain a sub-image region of interest (SROI) defined by (i) a position of the SROI within the sub-image, and (ii) a symbology category encompassing a plurality of barcode symbologies; generate a region of interest (ROI) from the SROI by converting the position of the SROI in the sub-image to a position in the image, the ROI defined by the position in the image and the symbology category; provide the ROIs to a barcode decoder; obtain a further SROI having a position within a further sub-image; and generate the ROI from the SROI by: identifying SROIs with contiguous positions in the image, and merging the SROIs with contiguous positions.

8. The computing device of claim 7, wherein the image pre-processor is further configured, prior to partitioning the image into the sub-images, to: determine a number of the sub-images based on a target size of the sub-image.

9. The computing device of claim 8, wherein the target size of the sub-image defines a pixel count; and wherein the image pre-processor is configured to determine the number of the sub-images by: dividing a pixel count of the image by the pixel count defined by the target size of the sub-image.

10. The computing device of claim 7, wherein the symbology category includes one of a first category corresponding to one-dimensional barcodes, and a second category corresponding to two-dimensional barcodes.

11. The computing device of claim 7, wherein the barcode decoder is configured to: select a set of barcode symbologies according to the symbology category corresponding to the ROI; determine which of the symbologies matches the ROI; and decode data from the ROI based on the symbology matching the ROI.

12. The computing device of claim 11, further comprising a display, wherein the barcode decoder is further configured to: control the display to present the position of the ROI and a visual indicator of a decoding status; determine a three-dimensional location corresponding to the ROI; determine, based on the three-dimensional locations, that the ROI was detected in a threshold number of previous images and has not been decoded; and update the visual indicator associated with the ROI.
